



Fuel on the go

NASA is looking into bionanotech that will convert plant and human waste into fuel for long distance space programmes.



And now, Biohackers

Renegade scientists are now manufacturing mutant bacteria strains in homebrew labs using videos on the internet. More at <http://snipurl.com/ql1xw>.

Feature

becomes significantly cleaner and cheaper. Bleaching the pulp to create white paper can be achieved with enzymes called xylanases, instead of the usual chlorine-containing chemicals, reducing the need to treat the waste water.

• Plastic industry

Plastics have revolutionised our lives, but their recalcitrance to degradation have made them ecologically unacceptable. Bioplastics offer the best of both worlds. These are different from normal plastics in two ways – the raw material and their molecular makeup. Normal plastics use petroleum as the main raw material, and their complex molecular structure makes them resistant to decomposition by micro organisms. Bioplastics, on the other hand, use starch as the raw material – starch is abundant in plants such as corn, potato, etc., which are renewable and easier to obtain. Also, their molecular structure is more simple making them more biodegradable. Just as with normal plastics, there are many types of bioplastics, all with different molecular structures. Polylactic acid (PLA) is a popular bioplastic material. This has its starting material lactic acid that is produced by the enzymatic breakdown of starch. Bioplastics can be moulded into different shapes and share the same utility possibilities as regular plastics.

• Bioremediation

This is the reclaiming of any area that has been polluted beyond human/animal inhabitability, using biological agents. Take the case of cleaning up oil spills. The BT answer to dealing with this challenge is to spray the slick with bacteria that prefer oily environments. These bacteria break up the oil and in the process release carbon dioxide and water, as part of their normal metabolism. The environmental impact of the cleaning up operation is almost nil (some chemicals need to be added to encourage the bacteria to speed up the process), since there are no by-products that are unsafe. Oil slicks are accidental in nature, but normal petroleum extraction and delivery activities

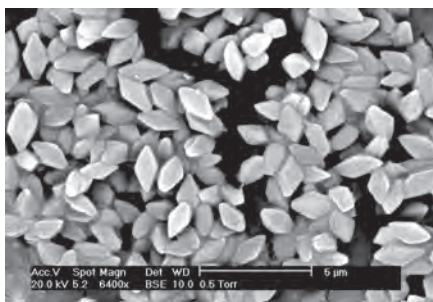
involve leakages and spills that taint the vicinity. MOs are also used in these areas to clean up the mess.

• Biofuels

These are the subject of a lot of public attention, especially after petroleum prices skyrocketed in the recent past, and also due to concern about global warming.

Ethanol can be used as an additive to normal petrol – a mixture of 85 per cent ethanol and 15 per cent petrol has been found usable with minimal engine modification

The Jatropha plant has been in the news for being a source of biodiesel – oil derived by crushing its seed is mixed with alcohol to make biodiesel – which is an ideal substitute to diesel. BT offers other avenues of creating substitute fuels such as ethanol (ethyl alcohol), which can also be used in modern vehicles that otherwise run on petrol. The core process involves using appropriate enzymes to break down plant matter to create the ethanol. The plant matter can be plant products such as corn, or sugarcane, or any plant waste product or even wood from trees – the enzyme needed to perform the break down changes with the source. Ethanol can be used as an additive to normal petrol – a mixture of 85 per cent ethanol and 15 per cent petrol has been found usable with minimal engine modification. Biogas/"gobar" gas plants are quite well known here in India, and BT plays a role in



BT: An enemy of an enemy

the process of using bacteria to break down dung/"gobar" into methane which can then be used as burning fuel.

• Biomining

MOs are used to aid the process of extracting metal from its ore, mostly sulphide ores. This process is used in the extraction of metals such as copper, nickel, gold, zinc, etc. At present, about 25 per cent of the total copper mined uses MOs to extract impurities from the ore before further processing. Gold miners also benefit from using MOs in the extraction process. Algae are used to improve the oil output from oil wells. Once injected into the oil well, algae, as a normal by-product of their metabolism, produce carbon dioxide gas which displaces the petroleum and facilitates recovery.



Biodegradable plastic cutlery is a commercially available product


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